

Lecture 2: Components of Social Media Platforms and Web Scraping

IS 618: Social Media Data Analysis
Indira Sen, FSS 2026

Recap: Why do we study society and social phenomena?

- ❖ To learn about society
 - How do sexist attitudes propagate?
 - Do disease outbreaks also affect mental health?
- ❖ To design policies and interventions for addressing societal problems
 - Hate speech
 - Polarization

How do we study society and social phenomena?

- ❖ To learn about society
 - How do sexist attitudes propagate?
 - Do disease outbreaks also affect mental health?



Questionnaire
Design



Solicitation of
Responses

Surveys

- Declining response rates
- Reporting biases
- ...

- ❖ To design policies and interventions for addressing societal problems
 - Hate speech
 - Polarization



Manual
analysis

- Emotional and psychological trauma
- Intractable at scale
- ...

The image shows a screenshot of a tweet from Jon Snow (@kingofthenorth) dated June 1, 2021. The tweet text is: "*yawn* Im sorry but women cannot drive, call me sexist or whatever but it is true." The phrase "women cannot drive," is highlighted with a red box. A blue box labeled "Input: Text to be labeled" has an arrow pointing to the highlighted text. A red box labeled "Output: Sexist; stereotypes" has an arrow pointing from the highlighted text. The tweet also shows 64 Retweets, 32 Quote Tweets, and 422 Likes.

Jon Snow
@kingofthenorth

yawn Im sorry but women cannot drive, call me sexist or whatever but it is true."

12:00 PM · Jun 1, 2021

64 Retweets 32 Quote Tweets 422 Likes

Input: Text to be labeled

Output: Sexist; stereotypes

Enter: web and social media data
+ computational methods

Computational Social Science

David Lazer,¹ Alex Pentland,² Lada Adamic,³ Sinan Aral,^{2,4} Albert-László Barabási,⁵ Devon Brewer,⁶ Nicholas Christakis,¹ Noshir Contractor,⁷ James Fowler,⁸ Myron Gutmann,³ Tony Jebara,⁹ Gary King,¹ Michael Macy,¹⁰ Deb Roy,² Marshall Van Alstyne^{2,11}

A field is emerging that leverages the capacity to collect and analyze data at a scale that may reveal patterns of individual and group behaviors.

Download

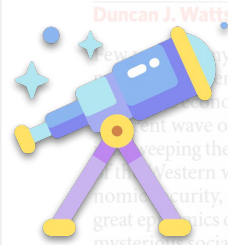
We live life in the network. We check our e-mails regularly, make mobile phone calls from almost any location, swipe transit cards to use public transportation, and make purchases with credit cards. Our movements in public places may be captured by video cameras, and our medical records stored as digital files. We may post blog entries accessible to anyone, or maintain friendships through online social networks. Each of these transactions leaves digital traces that can be compiled into comprehensive pictures of both individual and group behavior, with the potential to transform our understanding of our lives, organizations, and societies.

The capacity to collect and analyze massive amounts of data has transformed such fields as biology and physics. But the emergence of a data-driven "computational social science" has been much slower. Leading journals in economics, sociology, and political science show little evidence of this field. But computational social science is occurring—in Internet companies such as Google and Yahoo, and in govern-

NATURE | Vol 445 | 1 February 2007

A twenty-first century science

If handled appropriately, data about Internet-based communication and interactivity could revolutionize our understanding of collective human behaviour.



Duncan J. Watts

New research suggests that many of the major trends of the twenty-first century will be economic and social. The current wave of religious and political movements (keeping the Islamic world in the Western world), to economic security, global warming and the great epiphanies of our times, powerful yet mysterious social forces come into play.

But few readers of *Nature* would consider social science to be the science of the twenty-first century. Although economics, sociology, political science and anthropology have produced a plethora of findings regarding human social behaviour, they

self-reports from participants, which suffer from cognitive biases, errors of perception and framing, and ambitious

Digital Traces + computational methods are the ***Telescope*** for the Social Sciences

actions of millions of people at a resolution that is sensitive to effects at the level of the individual. Meanwhile, ever-faster computers permit us to simulate large networks of social interactions. The result has been tremendous interest in social networks: thousands of papers and a growing number of books have been published in less than a decade, leading some to

framework of collective social dynamics. People do not just interact: their interactions have consequences for the choices they, and others, make.

Studies that combine all these features are currently beyond the state of the art, but two of my group's recent projects indicate tentative progress. The first used the anonymized e-mail logs of a university community of around 40,000 people to track daily network evolution over a year as a function of existing network structure, shared activities (such as classes) and individual attributes. Dynamic data of this type may shed light on the relative roles of struc-

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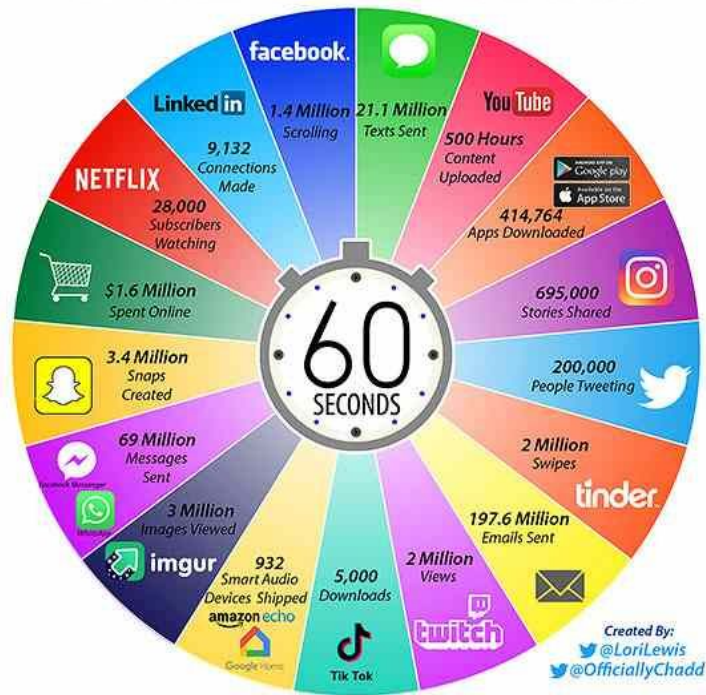
Data from the blogosphere. Shown is a link structure within a community of political blogs (from 2004), where red nodes indicate conservative blogs, and blue liberal. Orange links go from liberal to conservative, and purple ones from conservative to liberal. The size of each blog reflects the number of other blogs that link to it. [Reproduced from (8) with permission from the Association for Computing Machinery]

Reference:

[Computational social science](#)
[A Twenty-first Century Science](#)

What is social media?

2021 *This Is What Happens In An Internet Minute*



What does social media have to do with (social) science?

Social Media can advance Social Science in the following ways:

1. Use Social Media to learn *about* society
2. Study Social Media's *impact on* society

Social Media Data *complements* Traditional Data Sources

Article

The Price of Power: Power Seeking and Backlash Against Female Politicians

Tyler G. Okimoto¹ and Victoria L. Brescoll¹

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Psychology Bulletin
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DOI: 10.1177/0146167210371949
http://pspb.sagepub.com


Abstract

Two experimental studies examined the effect of power-seeking intentions on backlash toward women in political office. It was hypothesized that a female politician's career progress may be hindered by the belief that she seeks power, as this desire may violate prescribed communal expectations for women and thereby elicit interpersonal penalties. Results suggested that voting preferences for female candidates were negatively influenced by her power-seeking intentions (actual or perceived) but that preferences for male candidates were unaffected by power-seeking intentions. These differential reactions were partly explained by the perceived lack of communality implied by women's power-seeking intentions, resulting in lower perceived competence and feelings of moral outrage. The presence of moral-emotional reactions suggests that backlash arises from the violation of communal prescriptions rather than normative deviations more generally. These findings illuminate one potential source of gender bias in politics.

Keywords

gender stereotypes, backlash, power, politics, intention, moral outrage

Okimoto et al., [The Price of Power: Power Seeking and Backlash Against Female Politicians](#)

- Uses a survey experiment where people are shown fictional politicians with different traits and asked to choose whom they prefer

Full & Short Papers

WebSci '19, June 30–July 3, 2019, Boston, MA, USA

As the Tweet, so the Reply? Gender Bias in Digital Communication with Politicians

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ABSTRACT

This study investigates gender bias in political interactions on digital platforms by considering how politicians present themselves on Twitter and how they are approached by others. Incorporating social identity theory, we use dictionary analyses to detect biases in individual tweets connected to the German federal elections in 2017. Besides sentiment analysis, we introduce a new measure of personal- vs. job-related content in text data, that is validated with structural topic models. Our results indicate that politicians' communication on Twitter is driven by party identity rather than gender. However, we find systematic gender differences in tweets

1 INTRODUCTION

"Why do you even care for the #BER [Berlin airport] opening? Because you have to choose your clothes for the opening ceremony???"

This tweet directed at a female German politician resembles an often observed and reported phenomenon in our digital society: digital communication is driven by gender stereotypes rather than job-related content. This, in turn, can lead to bias and discrimination towards female professionals. Building on insights from research on social identities [60, 61] as well as gender biases and gender roles in

Mertens et al., [As the Tweet, so the Reply? Gender Bias in Digital Communication with Politicians](#)

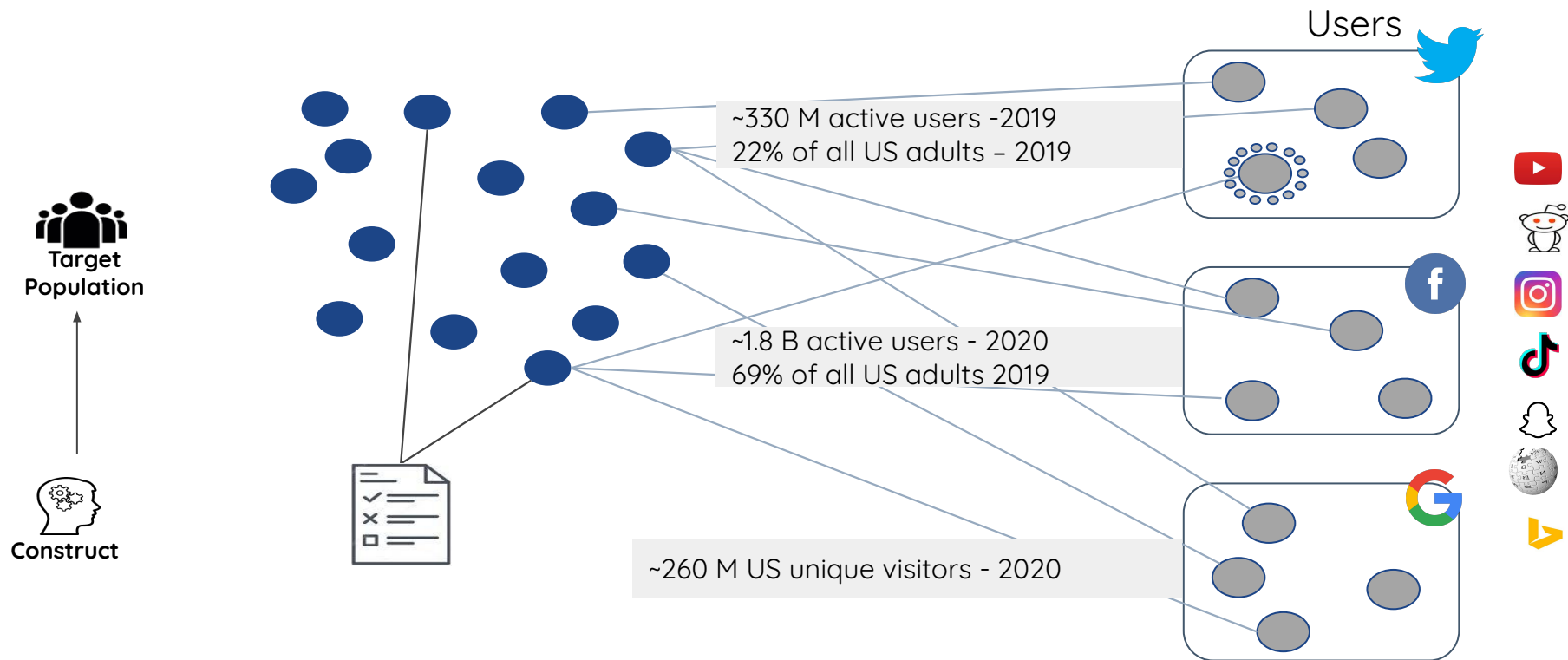
- Uses tweets mentioning real-life politicians and computationally assesses stereotypes in those tweets

Understanding Social Phenomena

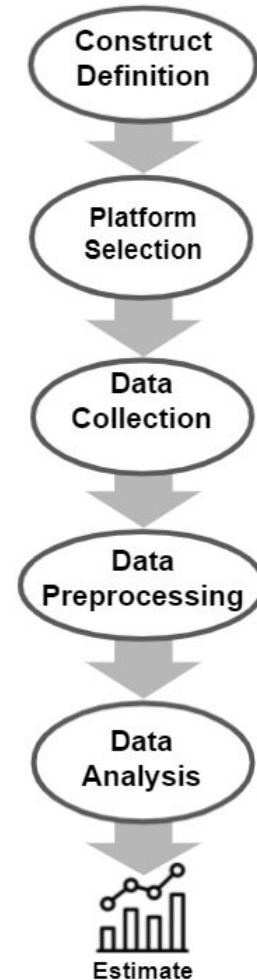


A typical research pipeline with digital trace data

A typical research design with digital traces



Prototypical Pipeline of Research with Social Media Data



Prototypical Pipeline: Data Collection

collect data potentially containing tangible signals regarding our construct

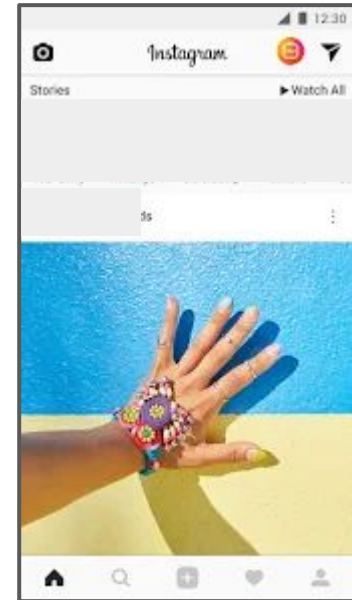
Content



videos



text



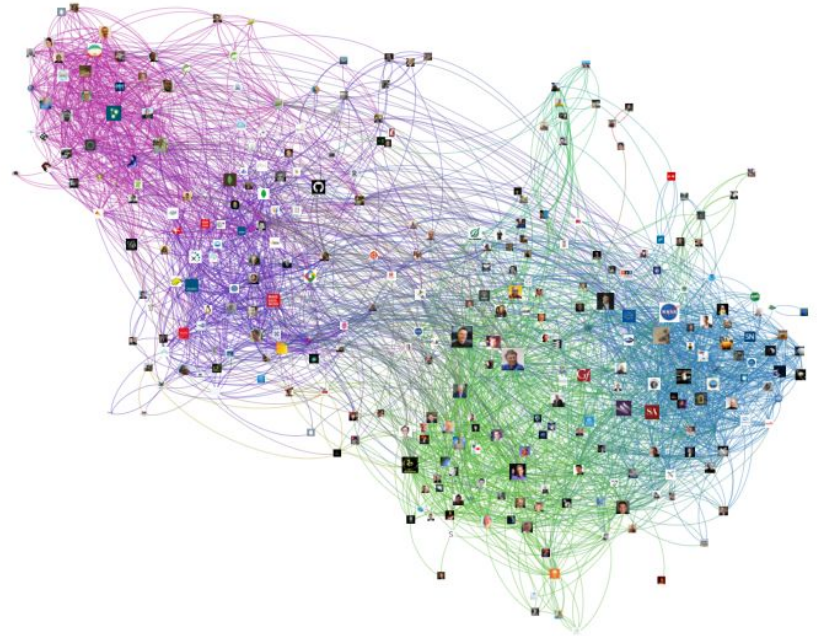
images

Prototypical Pipeline: Data Collection

collect data potentially containing tangible signals regarding our construct

Other data:

- **Connections** (friends, followers, links/URLs)
- **Actions** (likes, favs, comments)
- **Aggregate data** (views, shares, top searches)
- **User profiles** (with metadata)



Prototypical Pipeline: Data Collection

Source types:

- Application Programming Interfaces (API)
- Web scraping
- Web tracking
- Data donations
- Reuse existing datasets (data archives)
- Official sale
- Use 3rd party services (e.g. Pushsift, Crimson Hexagon)

A rough list of (inter-connected) methods

- Network analysis
- Content analysis
 - Text analysis
 - Image and video analysis
- Human Computer Interaction ('social computing')
- Causal inference
- Combinations of two or more approaches

Studying Polarization with Twitter

- **Construct:** political polarization in the U.S during mass shootings
- **Platform:** Twitter
- **Data collected:** 4.4 Million tweets with keywords related to mass shootings collected with the Twitter API
- **Analysis:** many methods of content analysis including topic modeling
- **Result:** Democrats and Republicans focus on different aspects during mass shootings

Demszky et al., [Analyzing Polarization in Social Media: Method and Application to Tweets on 21 Mass Shootings](#)

Analyzing Polarization in Social Media: Method and Application to Tweets on 21 Mass Shootings

Dorottya Demszy¹ Nikhil Garg¹ Rob Voigt¹ James Zou¹

Matthew Gentzkow¹ Jesse Shapiro² Dan Jurafsky¹

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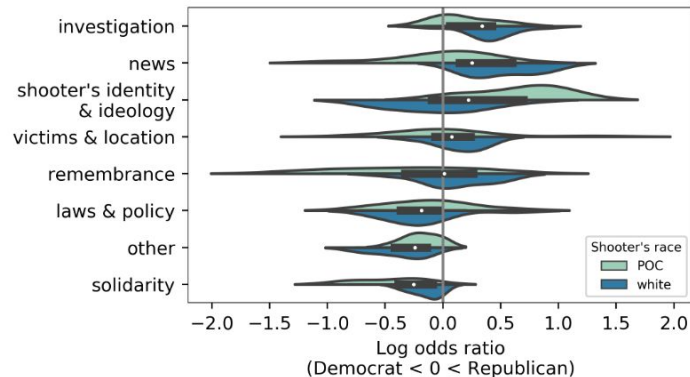


Figure 7: The plot shows the kernel density of the partisan log odds ratios of each topic (one observation per event). The white points show the median and the black rectangles the interquartile range across events.

(Bakshy et al., 2015). Prior to this, e.g., that polarized messages to be shared (Zafar et al., 2012); however, we lack understanding of the many ways that this is instantiated linguistically. We provide a more comprehensive framework for studying linguistic aspects of polarization by looking at topic choice, framing, and locutionary force.

Subjects of polarization by study-

Studying Politics with Tiktok vs. Youtube

- **Construct:** Political content
- **Platform:** TikTok and Youtube
- **Data collected:** 11K TikTok accounts who primarily post about politics and 1,712 political YouTube channels
- **Analysis:** descriptive summary of views, comments, and playbacks
- **Result:** Videos reach beyond the network more easily on Tiktok

Fifteen Seconds of Fame: TikTok and the Supply Side of Social Video

Benjamin Guinaudeau
University of Konstanz

Kevin Munger
Pennsylvania State University

Fabio Votta
University of Amsterdam

Abstract

TikTok has rapidly developed from a punchline for jokes about “kids these days” into a formidable force in American politics. The speed of this development is unprecedented, even in the rapidly-changing world of digital politics. Through a combination of hashtag and snowball sampling, we identify 11,546 TikTok accounts who primarily post about politics, allowing us to analyze trends in the posting, viewing and commenting behavior on 1,998,642 tiktoks they have uploaded. We test a number of theories about how the unique combination of affordances on TikTok shapes how it is used for political communication. Compared to the dominant platform for political videos (YouTube) we find that a higher

Guinaudeau, Benjamin, Kevin Munger, and Fabio Votta.

["Fifteen seconds of fame: TikTok and the supply side of social video."](#)

The End of Theory: The Data Deluge Makes the **Scientific Method** Obsolete

By Chris Anderson 06.23.08

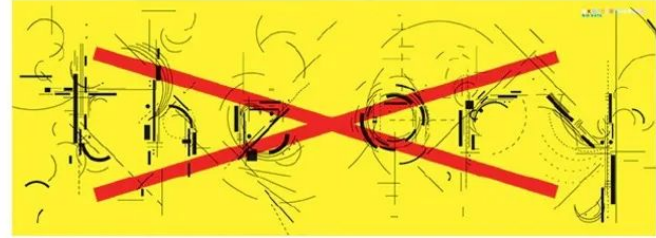


Illustration: Marian Bantjes

"All models are wrong, but some are useful."

So proclaimed statistician George Box 30 years ago, and he was right. But what choice did we have? Only models, from cosmological equations to theories of human behavior, seemed to be able to consistently, if imperfectly, explain the world around us. Until now. Today companies like Google, which have grown up in an era of massively abundant data, don't have to settle for wrong models. Indeed, they don't have to settle for models at all.

Sixty years ago, digital computers made information readable. Twenty years ago, the Internet made it reachable. Ten years ago, the first search engine crawlers made it a single database. Now Google and like-minded companies are sifting through the most measured age in history, treating this massive corpus as a laboratory of the human condition. They are the children of the Petabyte Age.

The Petabyte Age is different because more is different. Kilobytes were stored on floppy disks. Megabytes were stored on hard disks. Terabytes were stored in disk arrays. Petabytes are stored in the cloud. As we moved along that progression, we went from the folder analogy to the file cabinet analogy to the library analogy to — well, at petabytes we ran out of organizational analogies.

At the petabyte scale, information is not a matter of simple three- and four-dimensional taxonomy and order but of dimensionally agnostic statistics. It calls for an entirely different <https://www.wired.com/2008/06/pb-theory/> ^{e visualized in t for it later.}

For instance, Google conquered the advertising world with nothing more than applied mathematics. It didn't pretend to know anything about the culture and conventions of

Studying Social Media: No Big Deal?

Use case: Detecting the flu with digital traces

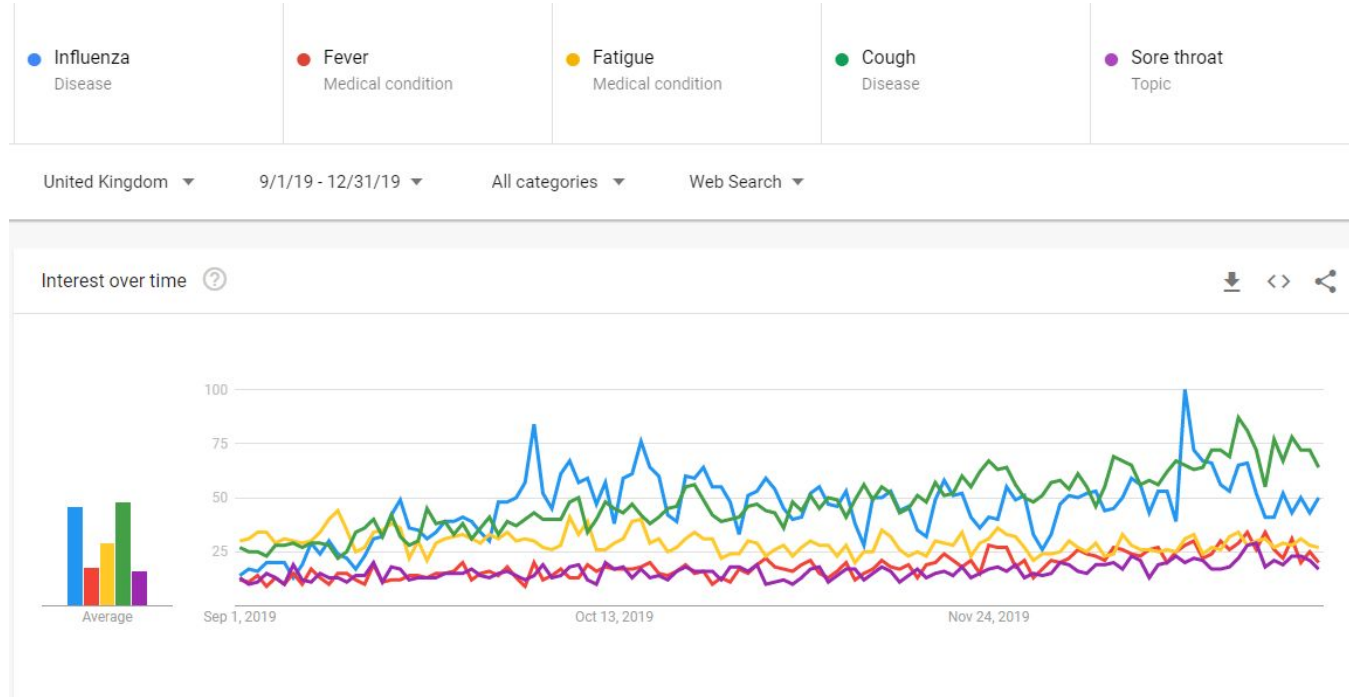
Google Flu (trends)



What proportion of US-Americans have the flu?

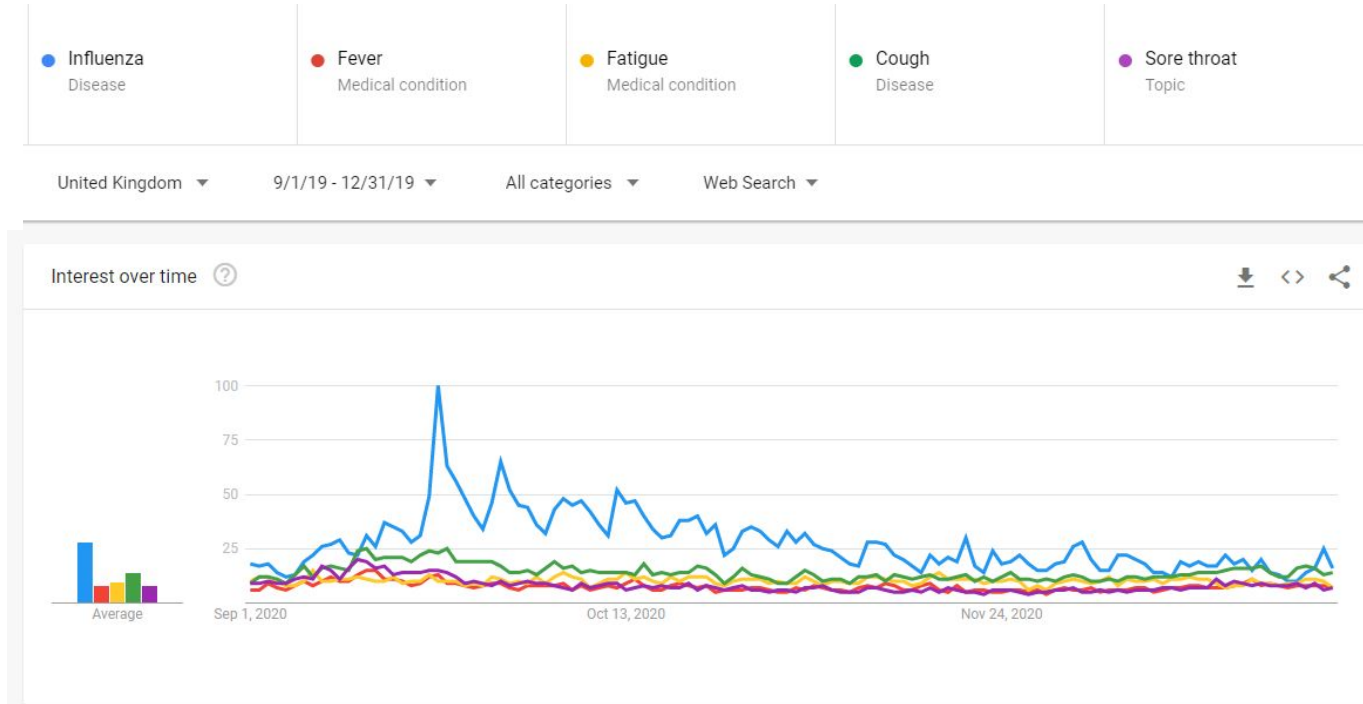
Approach

- Trends in Google searches related to influenza-like illnesses



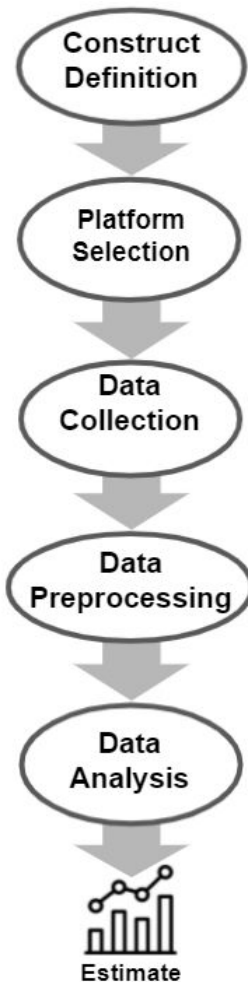
Approach

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Approach

- Construct: influenza-like illness (ILI)
- Target population(s): national and regional populations of the US
- Platform: Google web search
- Data collection: Logs 2003-2008, weekly, From 50 M. search terms, select those that best predict.
- Preprocessing: aggregated per each region, normalized by overall search activity, location via IP.
- Analysis: Regression where DV= ILI doctor visits, IV= ILI-related search fractions, for 1152 data points



Approach

□ Result

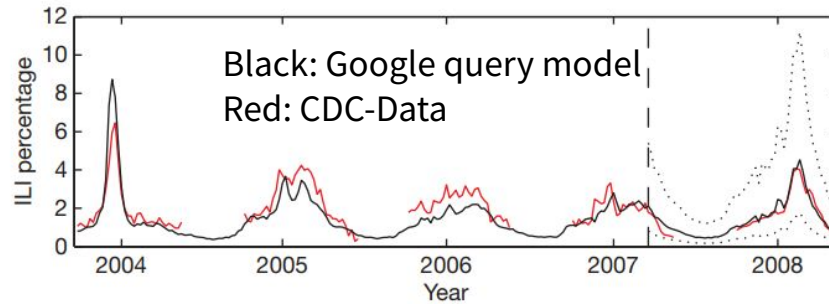
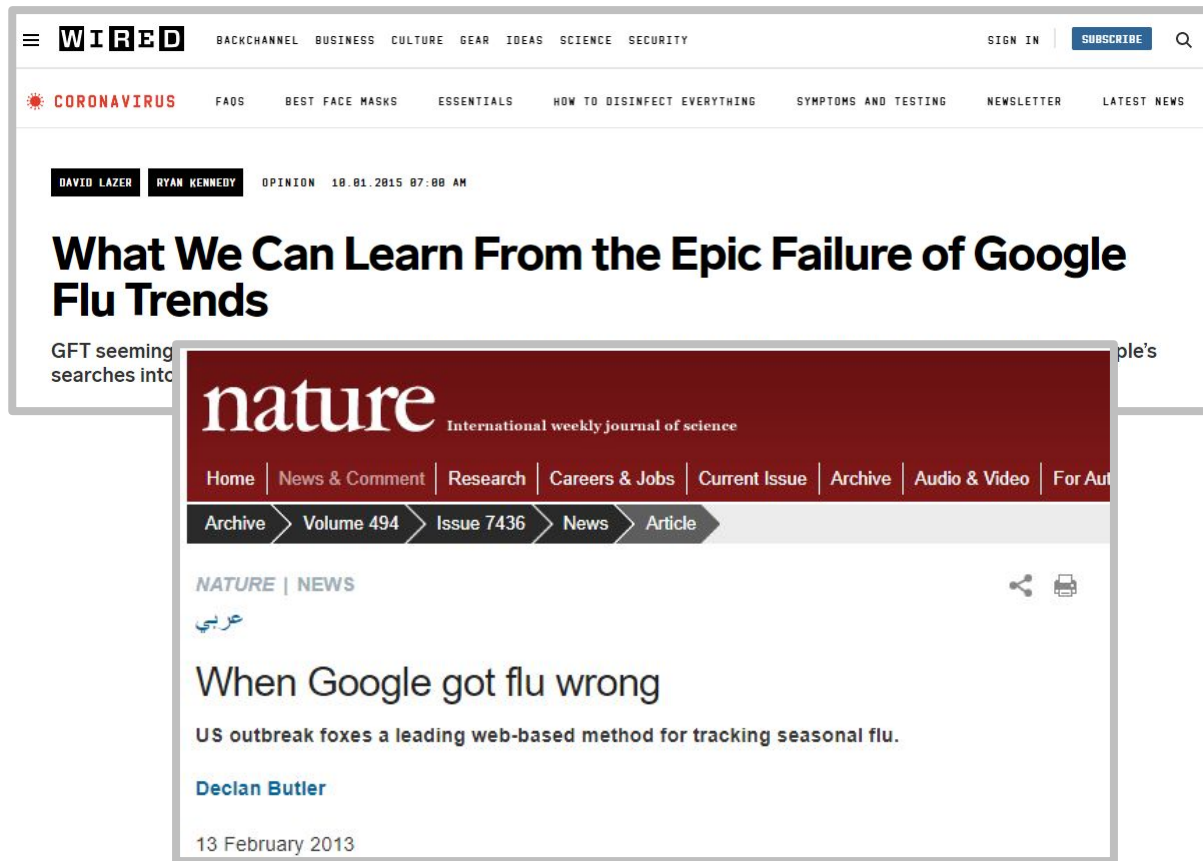


Figure 2 | A comparison of model estimates for the mid-2008 flu season (black) against CDC-reported ILI percentages (red). The model was fit over 128 points from 2004 to 2007, with a correlation of 0.96. The dotted line indicates 95% prediction intervals for New York, New Jersey and Pennsylvania.

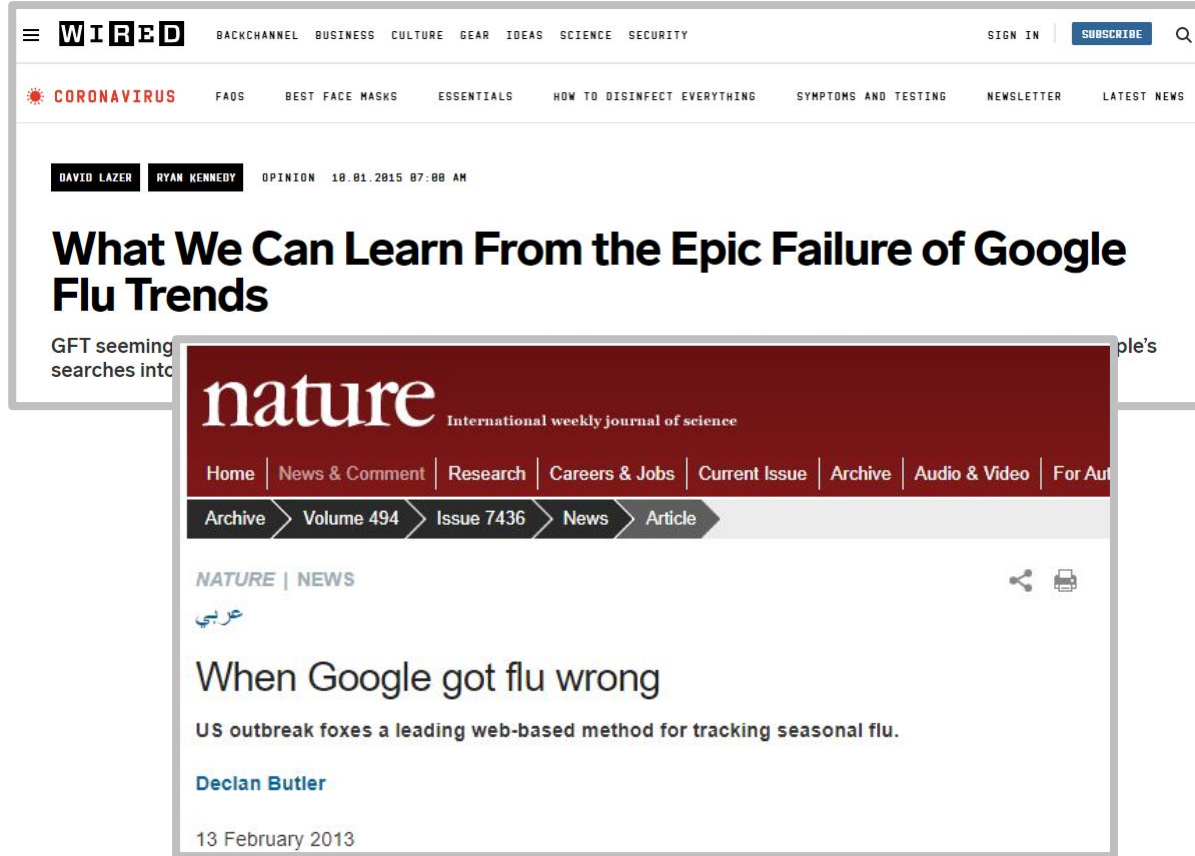
Problem solved

□ Now possible: Now- or fore-casting, fine-grained local detection, other countries

The fail

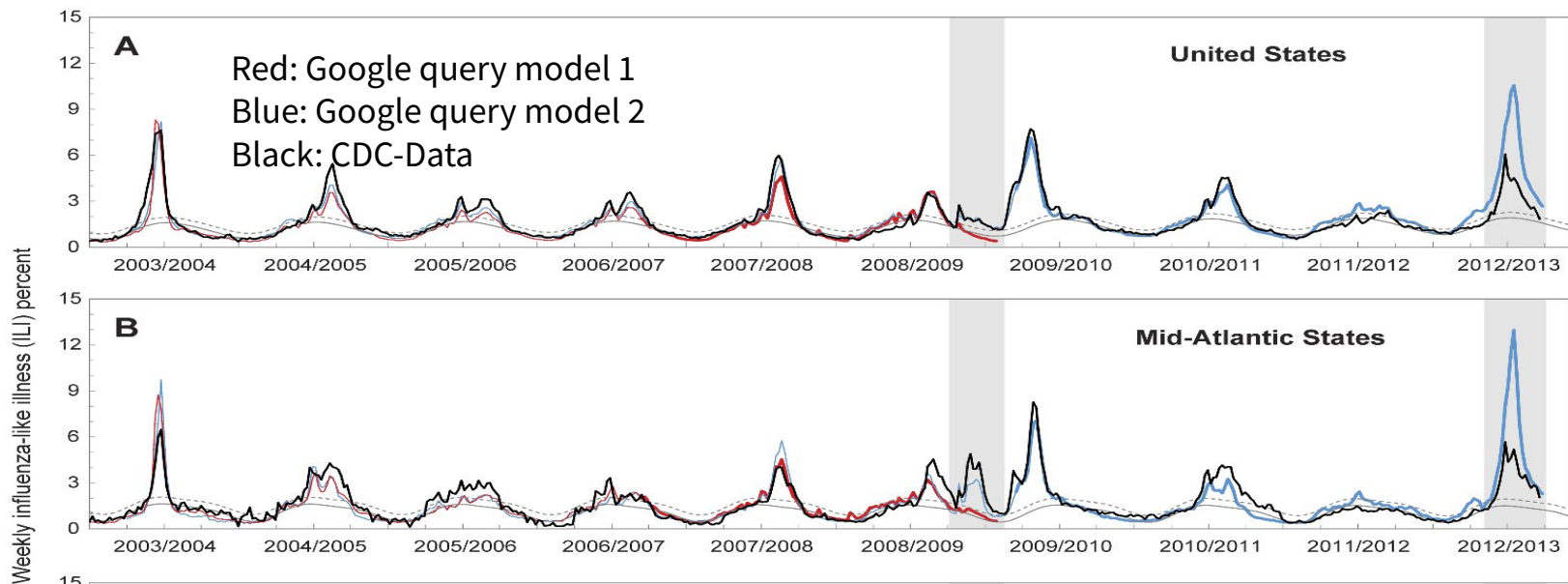


The fail



“In February 2013, Google Flu Trends (GFT) made headlines but not for a reason that Google executives or the creators of the flu tracking system would have hoped. Nature reported that GFT was predicting more than double the proportion of doctor visits for influenza-like illness (ILI) than the Centers for Disease Control and Prevention (CDC), which bases its estimates on surveillance reports from laboratories across the United States (1, 2).”

Overestimation



“part flue detector, part winter detector” □ errors are auto-correlated & direction and magnitude change with seasons

Where to be careful

“[...] there are enormous scientific possibilities in big data [but] the core challenge is that most big data that have received popular attention are not the output of instruments designed to produce valid and reliable data amenable for scientific analysis.”

Lazer, David, et al. "[The parable of Google Flu: traps in big data analysis.](#)" *Science* 343.6176 (2014)

Where to be careful

“[...] there are enormous scientific possibilities in big data [but] the core challenge is that most big data that have received popular attention are not the output of instruments designed to produce valid and reliable data amenable for scientific analysis.”

“**Big data hubris** is the often implicit assumption that big data are a substitute for, rather than a supplement to, traditional data collection and analysis”

Lazer, David, et al. "[The parable of Google Flu: traps in big data analysis.](#)" *Science* 343.6176 (2014)

Social Media Data *complements* Traditional Data Sources

Article

The Price of Power: Power Seeking and Backlash Against Female Politicians

Tyler G. Okimoto¹ and Victoria L. Brescoll¹

Abstract

Two experimental studies were hypothesized to show that women may violate political norms by expressing voting preferences that are not explained by their competence and/or the violation of communication source of gender

Keywords

gender stereotypes, power, intention, moral outrage

Personality and Social Psychology Bulletin
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sagepub.com/journalsPermissions.nav
DOI: 10.1177/0146167210371940
http://pspb.sagepub.com

Full & Short Papers

Psychological Science, June 30–July 3, 2019, Boston, MA, USA

Bias in Digital Communication with Politicians

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Social Media Data Analysis not necessarily superior to traditional data analysis.
But it's yet another methodology that helps you study society

The Price of Power: Power Seeking and Backlash Against Female Politicians

- Uses a survey experiment where people are shown fictional politicians with different traits

As the Tweet, so the Reply? Gender Bias in Digital Communication with Politicians

- Uses tweets mentioning real-life politicians and computationally assesses stereotypes in those tweets

Social Media Data Analysis

The Good 🥳

- Social Media provide a nice **complementary** data source to traditional social science data ([Lazer et al. 2009](#), [2020](#))
- The platforms themselves are an object of study
- Large scale data => an opportunity for data-driven approaches, like natural language processing (NLP)

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The Bad 😞

- Disproportionate focus on particular platforms / sources ([Tufekci 2014](#))
- Many representative biases ([Olteanu et al., 2021](#))
- Ad-hoc measurement models (“what are we *really* measuring?”)
- Need for a measurement theory ([Jungherr 2018](#))

Social Media Data Analysis

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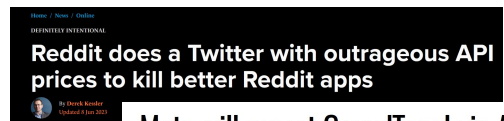
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The Ugly 🤢

- The API Apocalypse ([Freelon 2018](#); [Assenmacher et al., 2023](#))



Meta will sunset CrowdTangle in August

Advertisers and publishers will need to seek alternative platforms as CrowdTangle will no longer be available after August 14.

SCIENCE / TWITTER - X / TECH

Twitter just closed the book on academic research



/ Twitter was once an indispensable resource for academic research. That's changed under Elon Musk.



More on this next week

Beyond Methodological Questions: Ethics

- ❖ Research ethics practices are also still evolving
- ❖ Main focus on privacy
- ❖ Lack of informed consent
- ❖ “participants” not expecting research activities on their data
[[Fiesler & Proferes, 2018](#)]
- ❖ Different assumptions for different types of user groups (e.g. vulnerable groups)
- ❖ Potential starting point: [AoIR ethics guidelines](#)

Today:

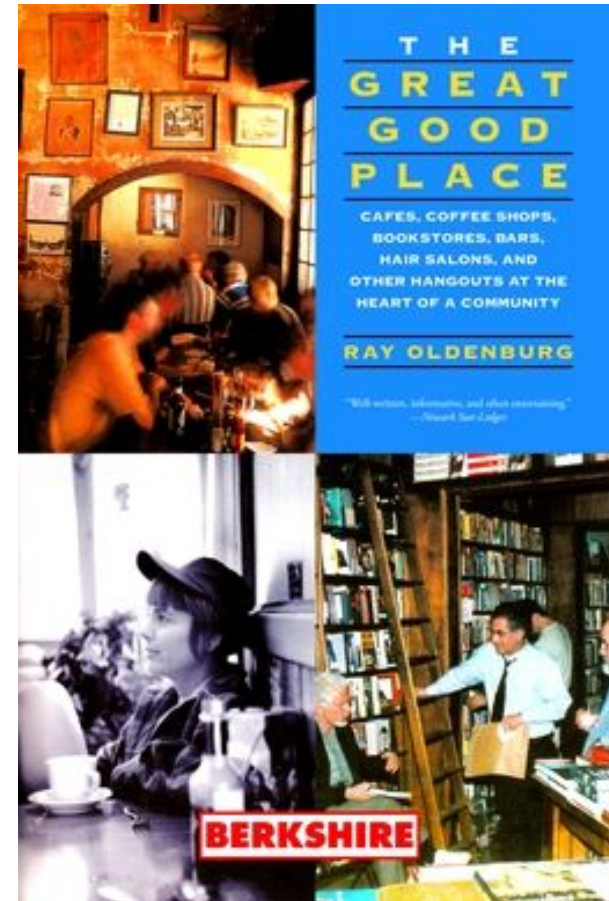
1. A Brief History of Social Media and the Elements of Modern Platforms
2. Data Collection 1: Web Scraping

Today:

- 1. A Brief History of Social Media and the Elements of Modern Platforms**
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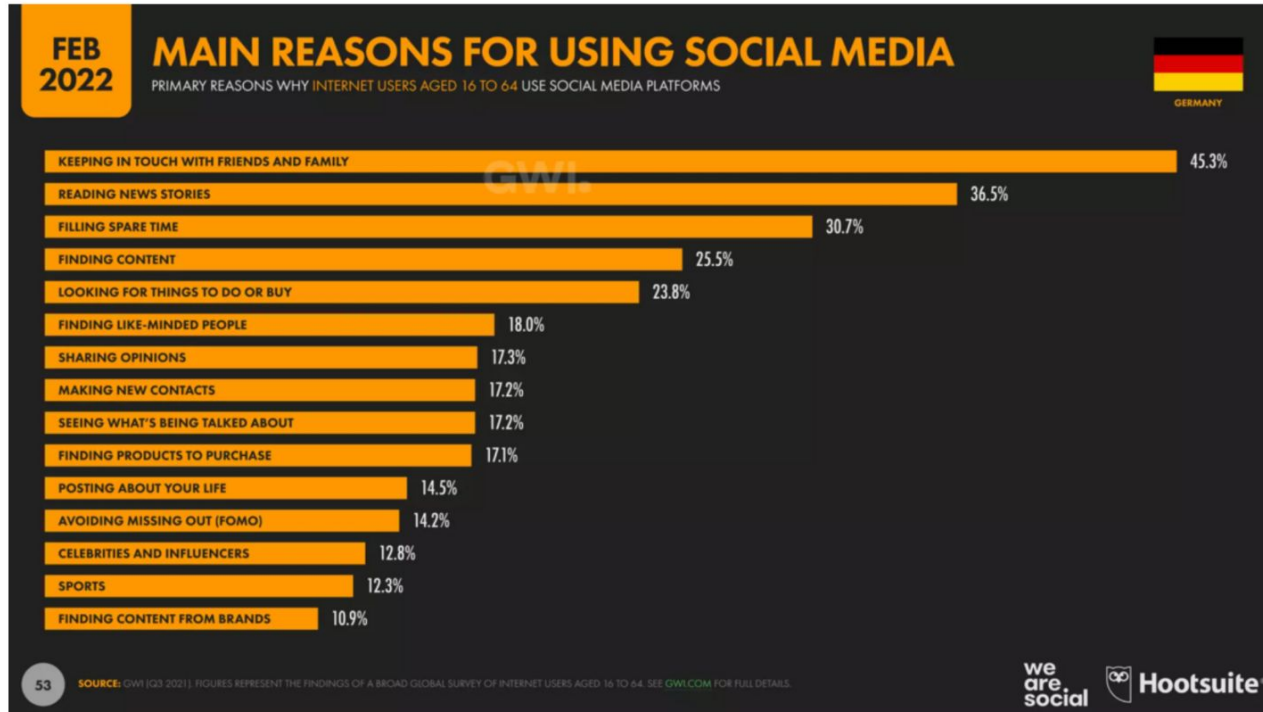
Why do we need social media?

- **Third Places:** “social surroundings that are separate from the two usual social environments of home (“first place”) and workplace (“second place”)”
 - Cafes, libraries, churches, vereins, ...
- Helps create a sense of community
- Physical third places have declined
- Social Media Platforms could be an alternative (but are they?) [[Bruckman 2022](#)]



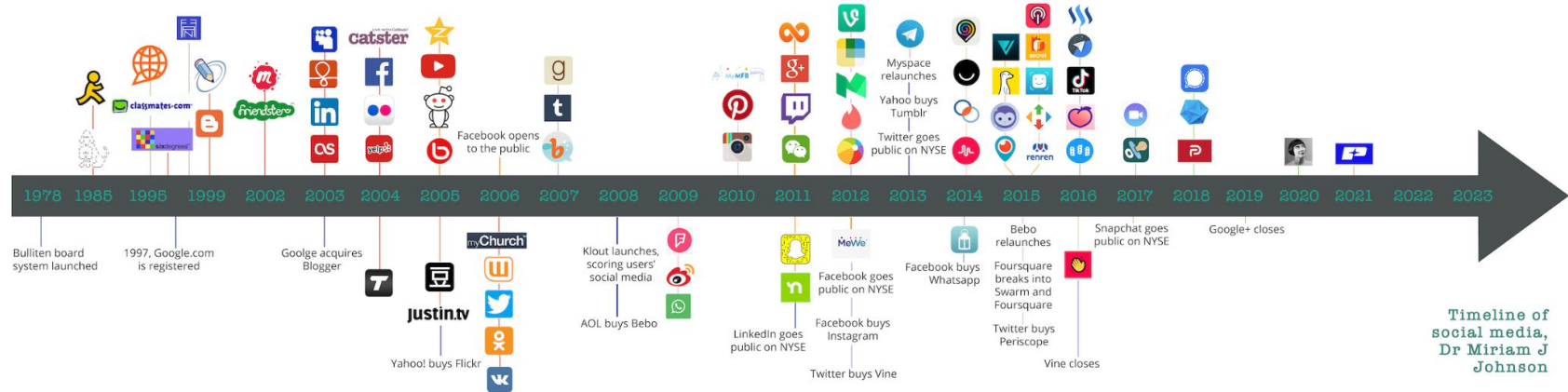
[Oldenburg, 1989](#)

Why do we people use social media?



<https://omr.com/en/reviews/contenthub/social-media-usage-germany>

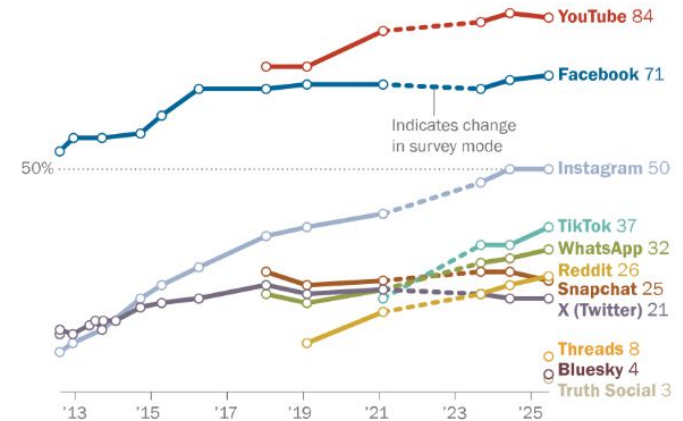
Evolution of Social Media Platforms



Evolution of Social Media Platforms

TikTok, Instagram, WhatsApp and Reddit have continued to gain users in recent years

% of U.S. adults who say they **ever** use the following



Note: The dotted line indicates a change in mode. Polls from 2012 to 2021 were conducted via phone. In 2023, the poll was conducted via web and mail. In 2024 and 2025, the poll was conducted via web, mail and phone. Refer to the topline for more information on how question wording varied over the years. Not all platforms were asked about in the full 2013-2025 timespan. Those who did not give an answer are not shown.

Source: Survey of U.S. adults conducted Feb. 5-June 18, 2025.

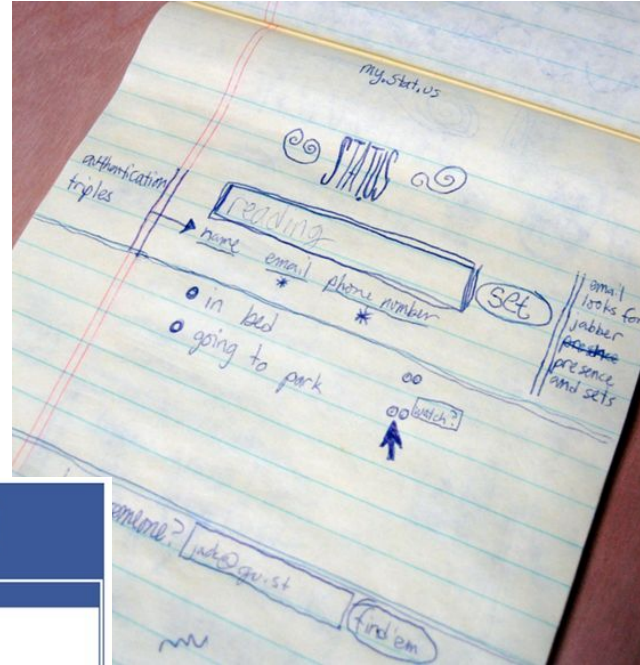
"Americans' Social Media Use 2025"

PEW RESEARCH CENTER

<https://www.pewresearch.org/internet/2025/11/20/americans-social-media-use-2025/>

Categorizing Different Social Media Platforms

- These platforms are complex with different, features, accessibilities, types of content, user bases, etc.
 - Not so easy to categorize
 - Moreover, platforms change considerably over time



<https://www.flickr.com/photos/jackdorsey/182613360>

Categorizing Different Social Media Platforms

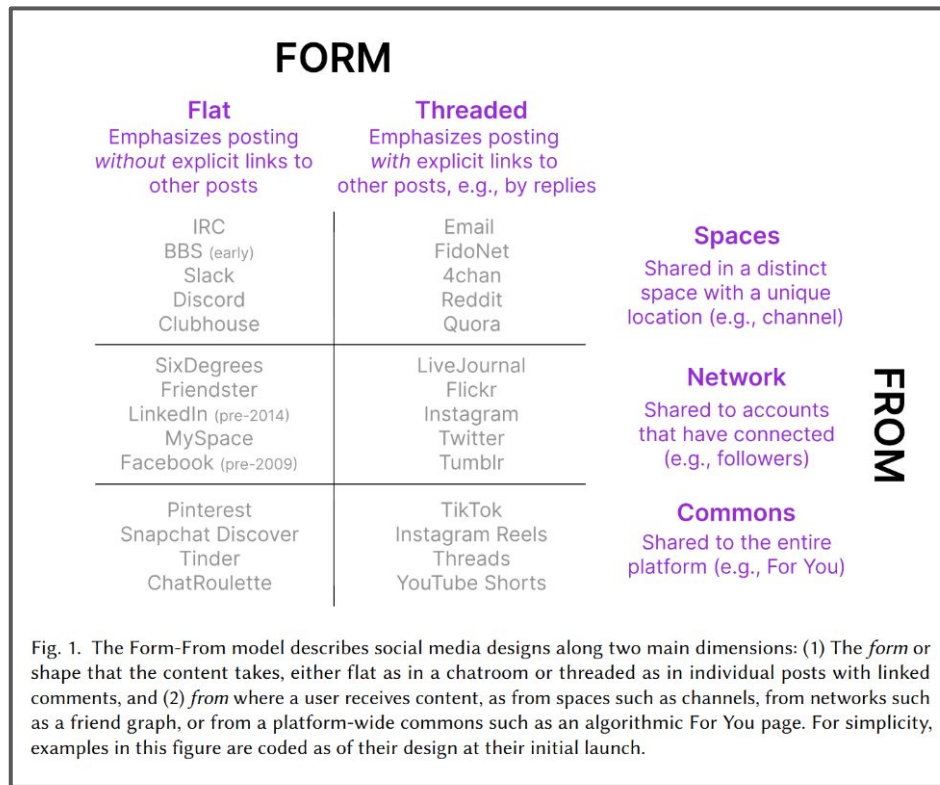
Some possible categorizations:

- **Ephemerality:** Snapchat (in early days) and stories vs. Facebook posts
- **Modality:** primarily text-based, e.g., Reddit, vs. Image-based e.g., Pinterest
- **Interaction type and timing:** e.g., BeReal
- **Moderation type:** Volunteer-run, e.g., Reddit vs. Commercial e.g., Facebook
- **Professional vs. personal use:** Facebook vs. LinkedIn
- **International or local:** Jappy, Pixnet
- **Value-based:** 'Free-speech' platforms like Truth Social

https://en.wikipedia.org/wiki/List_of_social_networking_services

Categorizing Different Social Media Platforms

- Form: the shape of the content:
 - **threaded** by emphasizing explicitly linking posts to each other
 - **flat** where the design emphasizes posts that are not explicitly linked to each other,
- From: content delivery
 - to the user from **spaces** (e.g., channels, groups, or rooms) where all members in the space receive it,
 - from (social) **networks** where each user independently curates their own network
 - a sample from the “**commons**” of all content on the platform?



FORM

Flat

Threaded

Spaces

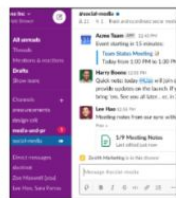
Network

Commons

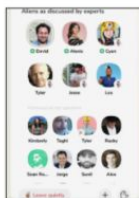
FROM



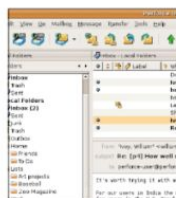
(a) Visual IRC



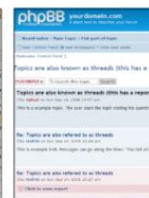
(b) Slack



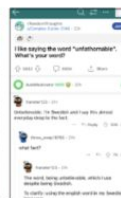
(c) Clubhouse



(d) Eudora email



(e) phpBB



(f) Reddit



(g) Friendster



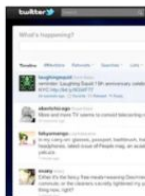
(h) MySpace



(i) early Facebook



(j) LiveJournal



(k) early Twitter



(l) Instagram



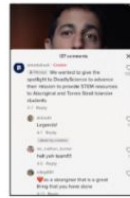
(m) Snapchat Discover



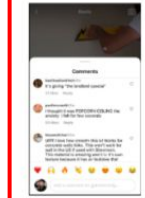
(n) Pinterest



(o) Tinder



(p) TikTok



(q) Instagram Reels



(r) Threads

Fig. 4. Examples of systems both old and current in each cell of the Form-From model.

The Elements of Modern Social Media Platforms

- The platform itself
- User base
 - Non-human users
 - User tiers
- Content
- Algorithms
 - Recommendation
 - Ranking
 - Moderation
- Advertisements

How do Platforms Make Money?

- Advertisements
- Subscriptions

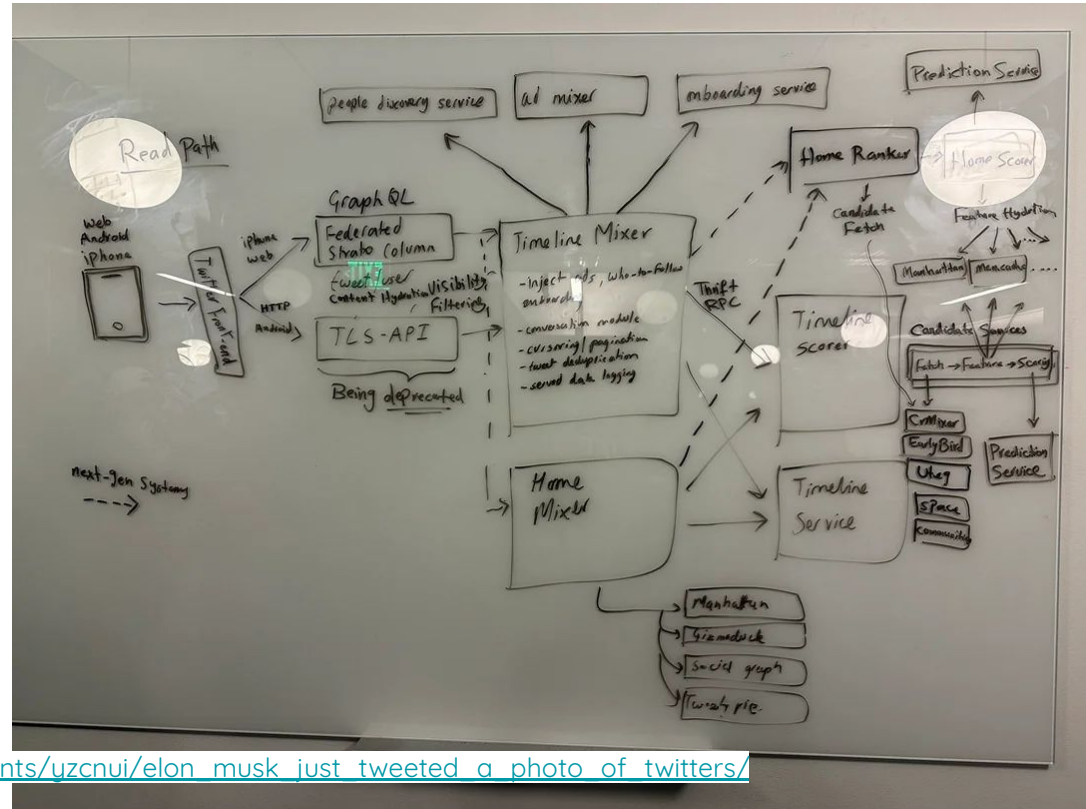
Class Activity: How would you design a new platform?

- Turn to person sitting beside you and discuss for the next 5-10 minutes:
- What type of content would it host?
 - Modality
 - Length
 - Format
 - ...
- What user base would you like to attract?
- How would you monetize it?
- What type of behavior would(n't) be allowed on the platform and how would you enforce it?

Today:

1. A Brief History of Social Media and the Elements of Modern Platforms
- 2. Data Collection 1: Web Scraping**

How is the Platform Itself Created?



https://www.reddit.com/r/webdev/comments/yzcnu/elon_musk_just_tweeted_a_photo_of_twitters/

How is the Platform Itself Created? The Software

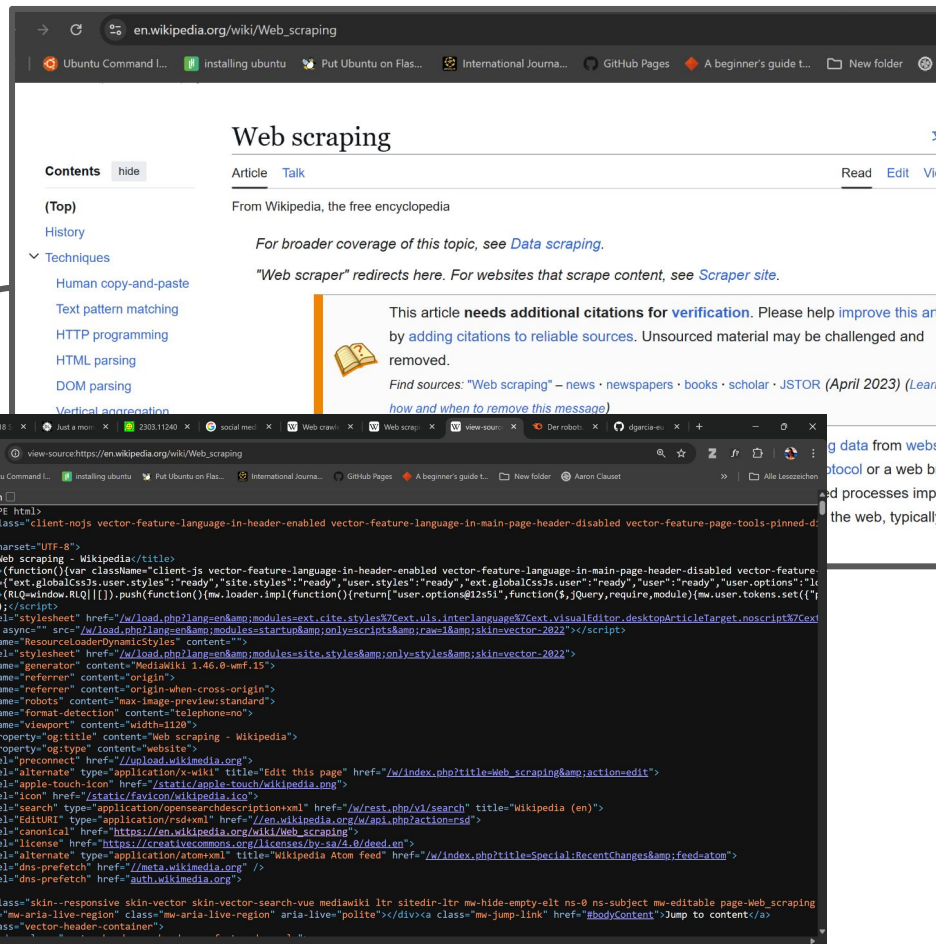
- A website and mobile apps where people interact - this is the "**front end**" that users see and tap on
- "**Back end**" handles all the actual work - storing posts, connecting friends, managing logins, etc.
- A **database** storing who all your users are, what they've posted, who's friends with whom, ...

How is the Platform Itself Created? The Hardware

- Data centers spread across different countries - you need these because millions of people are using your platform at once
- Servers that are always running, responding to requests like "show me my feed" or "upload this photo"

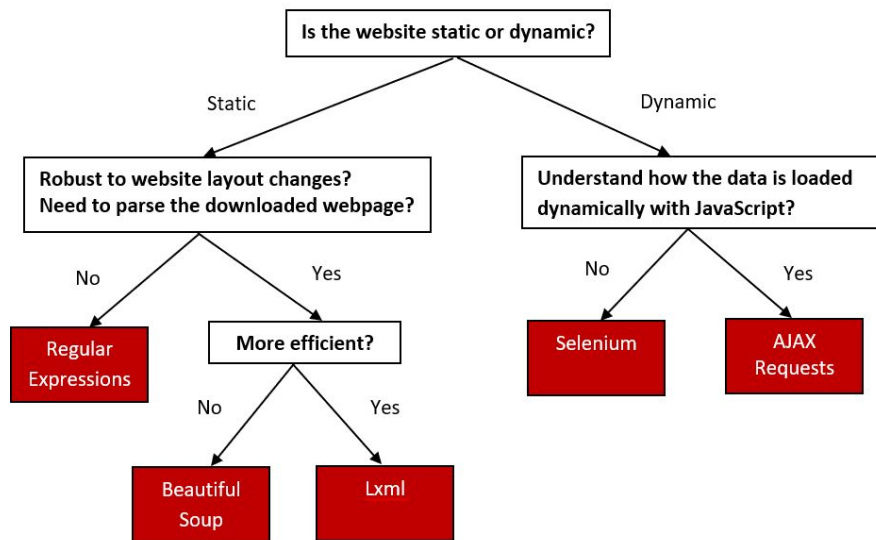
Web Scraping

- Webpages are created using text-based markup languages, e.g., HTML (Hypertext Markup Language)
- **Web scraping:** collecting the code and content used to create a webpage
- You can do this manually (e.g., by inspecting the source), but ideally through code



Web Scraping

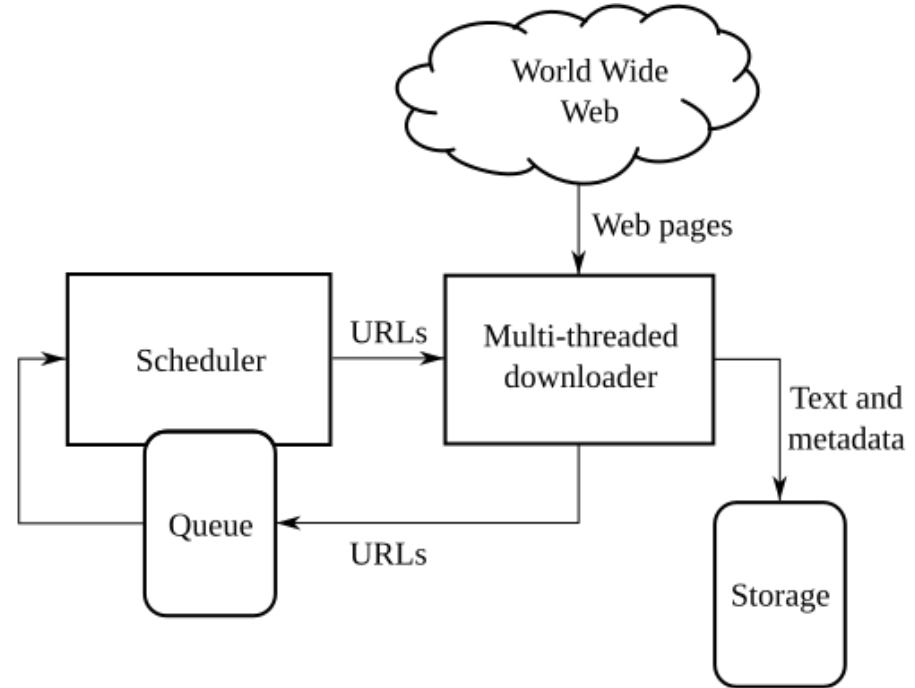
- For simple webpages, created with HTML and CSS, this is quite straightforward
- For dynamic webpages, more complicated
 - Most social media websites are dynamic — no two people see the same thing when they log in



<https://iqss.github.io/dss-webscrape/web-scraping-approaches.html>

Web Scraping vs. Web Crawling

- For collecting social media data, we don't just want to collect data from a single page
 - Multiple users' profiles from Twitter
 - Multiple subreddits on Reddit
- Thus you usually have a list of webpages that you need to scrape
- Therefore, you create crawlers that will do this process systematically



https://en.wikipedia.org/wiki/Web_crawler

What's in the Platform vs. What you can Collect

- **What you see vs. what you get:** With web scraping, whatever we see on the platform should be accessible
 - This is not true for APIs (as we will see in the next lecture)
- **What you don't even see:** The platform itself is probably storing much more data than they are showing you
 - E.g., the active browsers with Wikipedia pages open
 - Lurkers and lurking-related activity
 - Private and direct messages
 - ...

Is Web Scraping Legal?

- Depends on the region, but general rule of thumb: web scraping is allowed for non-commercial purposes and research
 - In Germany: the German Federal Court of Justice (Bundesgerichtshof) ruled that the scraping of publicly available data, including personal data, might be permissible under German data protection laws.
 - However, the scraped data must not be used for any illegal purposes, and that the scraping must not violate the rights of the data subjects.
- Can you make web scraped data widely available?
 - Conflict between privacy and reproducible research

Ethical Considerations

- Follow 'robots.txt' when possible
- Do not overload the website with requests
 - DoS: Denial of service attack
- Consider the pros and cons of research based on webscraping
 - Does it cause more benefit than harm?
 - Does it benefit or harm vulnerable people?



A screenshot of a web browser window displaying the content of the file `google.com/robots.txt`. The browser's address bar shows the URL `google.com/robots.txt`. The main content area displays the text of the robots.txt file, which includes instructions for search engine crawlers regarding which parts of the website are allowed or disallowed to be indexed.

```
User-agent: *  
Disallow: /search  
Allow: /search/about  
Allow: /search/static  
Allow: /search/howsearchworks  
Disallow: /sdch  
Disallow: /groups  
Disallow: /index.html?  
Disallow: /?  
Allow: /?hl=  
Disallow: /?hl=*  
Allow: /?hl=*&qws_rd=ssl$  
Disallow: /?hl=*&qws_rd=ssl
```

<https://de.semrush.com/blog/robots-txt-leitfaden/>

Use Case: The Truth Social Dataset

- Twitter clone that became popular after Donald Trump was banned from several social media platforms
- Interesting platform to study a population that maynot be active on 'mainstream' platforms

Truth Social Dataset

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Abstract

Formally announced to the public following former President Donald Trump's bans and suspensions from mainstream social networks in early 2022 after his role in the January 6 Capitol Riots, Truth Social was launched as an "alternative" social media platform that claims to be a refuge for free speech, offering a platform for those disaffected by the content moderation policies of the existing, mainstream social networks. The subsequent rise of Truth Social has been driven largely by hard-line supporters of the former president as well as those affected by the content moderation of other social networks. These distinct qualities combined with its status as the main mouthpiece of the former president positions Truth Social as a particularly influential social media platform and give rise to several research questions. However, outside of a handful of news reports, little is known about the new social media platform partially due to a lack of well-curated data. In the current work, we describe a dataset of over 823,000 posts to Truth Social and social network with over 454,000 distinct users. In addition to the dataset itself, we also present some basic analysis of its content, certain temporal features, and its network.

Introduction

The social media platform *Truth Social* was launched in February of 2022 about a year after the suspension of former United States President Donald Trump from Twitter, Facebook, and other social media platforms. Truth Social is largely stylized after Twitter where Tweets are instead called *Truths* and ReTweets are instead called *ReTruths*. Note that throughout the remainder of this paper we most commonly refer to *Truths* as *posts* in order to avoid confusion with the

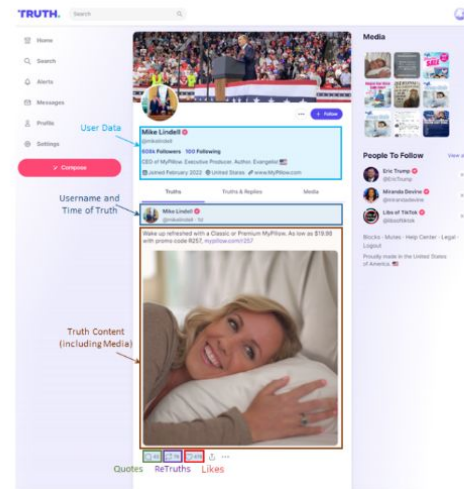


Figure 1: Annotated illustration of Truth Social Web Interface. The Web scraper extracted user data, and post information, including time, content (with links or other media), quotes, ReTruths, and likes.

Gerard, Patrick, Nicholas Botzer, and Tim Weninger. "[Truth social dataset](#)," Proceedings of the international AAAI conference on web and social media

Use Case: The Truth Social Dataset

- The authors use web crawling to collect data from Truth Social
- They use a ‘snowballing’ approach to collect data
- Do you think this approach affects the dataset quality in any way?

Dataset Collection Methodology

Collecting the posts and other activity data from Truth Social is not straightforward because the site does not provide a public API. Instead, we implemented a custom Web scraper and programmatically extracted the relevant context from Truth Social’s Web interface directly. This Web interface did not impose any crawling restrictions nor did it disallow any crawling with the `robots.txt` standard. Because no API was available the most straightforward way to collect posts was from each specific account.

We crawled the Truth Social one account at a time, starting with @realDonaldTrump and then iteratively in a breadth first manner over the followers of each account. Specifically, the crawling methodology proceeded as follows:

1. Collect information about the user’s follower count, following count, creation date.
2. Iterate through users following the user and users that the user follows, create an edge for each follower-followee relationship, and add that user to the breadth-first queue if that user has not been scraped in the past 14 days.
3. Extract the full set of available Truths posted by the user.

This crawl began on September 4, 2022 and continued until October 14, 2022. In that time, all content posted by 65,536 users was collected. The dataset, therefore, has the complete set of all posts for the visited users before September of 2022.

Summary and Next Week

- This session we learned:
 - The typical components of a social media platform
 - Data Collection I: Web scraping social media data
 - Use Case: Collecting data from Truth Social
- Next time:
 - Data Collection 2: Application Programming Interfaces (API)
 - Official and Unofficial APIs
 - Why Platforms offer APIs
 - Readings:
 - Background on APIs:
https://dgarcia-eu.github.io/SocialDataScience/Tutorials/2_SocialDynamics/027_APIs-Reddit/APIS-Reddit.html (you can ignore the part on R)
 - Murtfeldt, Ryan, et al. "[RIP Twitter API: A eulogy to its vast research contributions](#)," arXiv preprint arXiv:2404.07340 (2024).

Additional Resources

- Handbook on Social Media: Burgess, J., Marwick, A., & Poell, T. (2018). The sage handbook of social media. SAGE Publications Ltd,
<https://doi.org/10.4135/9781473984066>
- More on web scraping:
<https://www.scrapingdog.com/blog/best-python-web-scraping-libraries/>

Questions?